INCH-POUND

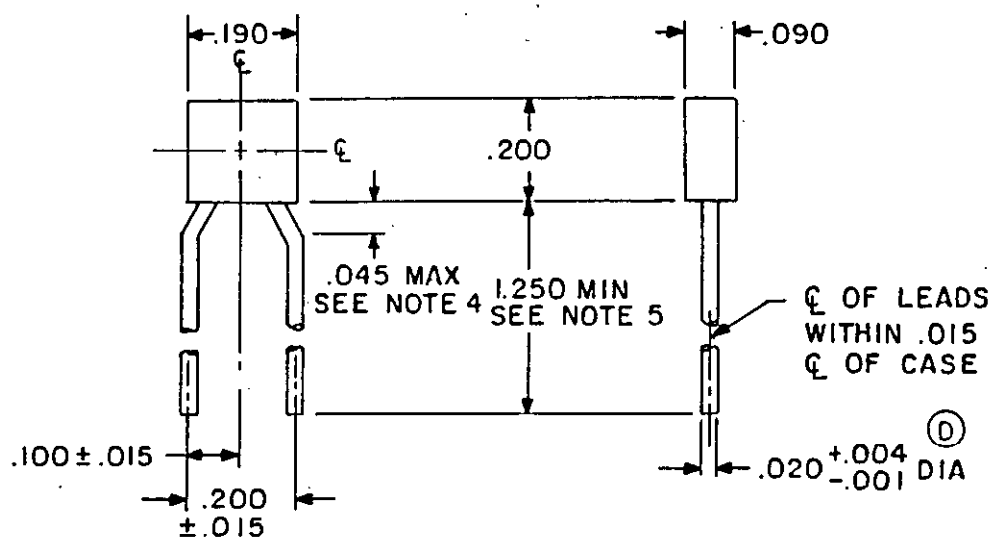
MIL-C-55514/8D
15 February 1989
SUPERSEDING
MIL-C-55514/8C
5 May 1986

MILITARY SPECIFICATION SHEET

# CAPACITORS, FIXED, METALLIZED PLASTIC DIELECTRIC, DC, IN NONMETAL CASES, ESTABLISHED RELIABILITY STYLES CFR12 AND CFR16

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-55514.




Style CFR12



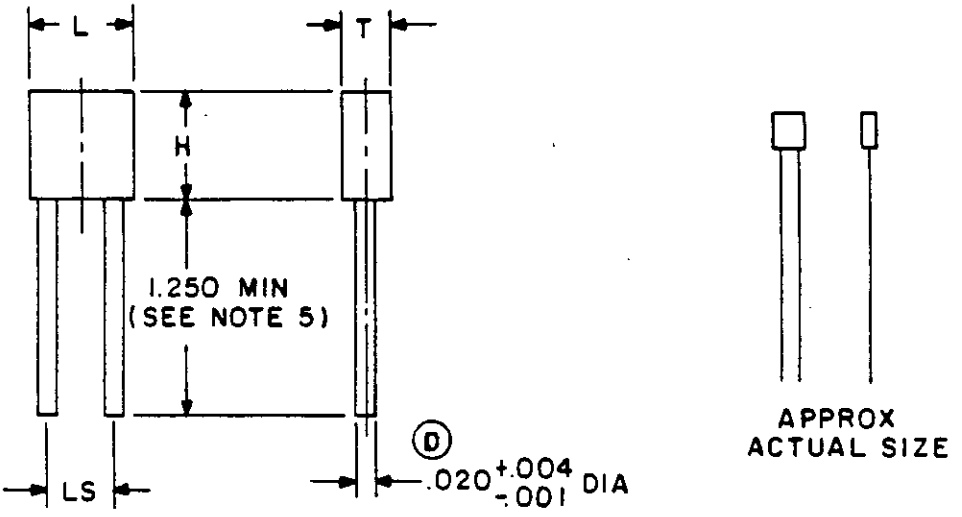
APPROX
ACTUAL SIZE

FIGURE 1. Dimensions and configuration.

Ⓓ denotes changes


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Style CFR16

| Inches | mm | Inches | mm |
|---|---|---|---|
| .001 | 0.03 | .090 | 2.29 |
| .004 | 0.10 | .100 | 2.54 |
| .015 | 0.38 | .190 | 4.83 |
| .020 | 0.51 | .200 | 5.08 |
| .045 | 1.14 | 1.250 | 31.75 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. For flush mounting .078 (1.98 mm) printed-circuit hole diameter is required to clear shoulder.
5. Lead length may be a minimum of one inch long for use in tape and reel packaging, when specified in the ordering data.

FIGURE 1. Dimensions and configuration - Continued.



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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1 and table I.

Case type: Molded or preformed.

Capacitance value: See table I.

Capacitance tolerance: See table I.

Operating temperature range: -55°C to +125°C.

Characteristic R: Will operate to +125°C when properly derated to 50 percent of its +85°C rated voltage.

Voltage rating: See table I.

Dissipation factor: Not more than 0.3 percent.

Marking: In accordance with MIL-C-55514 except that minimum marking as shown in the following example will be permitted. Full marking shall be included on the package.

| | Front | Reverse side | |
|---|---|---|---|
| Type designation | CFR 12<br>RRA<br>333 FM | J--<br>86 11 | J = JAN, and two digit manufacturer's code or symbol. Date code in accordance with MIL-C-55514. |

NOTE: Location of manufacturer's code or symbol is optional.

Substitutability for failure rate levels shall be as follows:

| Failure rate level | Will replace failure rate level |
|---|---|
| S | R, P, M |
| R | P, M |
| P | M |
| M | |

Substitutability for capacitance tolerance shall be as follows:

| Cap. tolerance | Will replace cap. tolerance |
|---|---|
| F | G, J, K |
| G | J, K |
| J | K |
| K | |

Substitutability for voltage shall be as follows:

| Volts dc | Will replace voltages |
|---|---|
| 250 | 200, 150, 100, 75, 50, 25 |
| 200 | 150, 100, 75, 50, 25 |
| 150 | 100, 75, 50, 25 |
| 100 | 75, 50, 25 |
| 75 | 50, 25 |
| 50 | 25 |
| 25 | |



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Ⓓ TABLE I. Styles CFR12 and CFR16 capacitor characteristics.

| Characteristic R capacitors | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Style CFR12 1/ | | | | Style CFR16 | | | | | | |
| Type designation 2/ | Capacitance | Rated voltage @ +85°C | Capacitance tolerance | Type designation 2/ | Capacitance | Rated voltage @ +85°C | Capacitance tolerance | T ±.010(0.25) | H ±.020(0.51) | L ±.010(0.25) | L.S. ±.015(0.38) |
| | µF | (volts) | | | µF | (volts) | | | | | |
| CFR12RRA333-- | .033 | 50 | F, G, J, K | CFR16RRA683-- | .068 | 50 | F, G, J, K | .095(2.41) | .245(6.22) | .295(7.49) | .200(5.08) |
| CFR12RRA393-- | .039 | 50 | F, G, J, K | CFR16RRA823-- | .082 | 50 | F, G, J, K | .095(2.41) | .245(6.22) | .295(7.49) | .200(5.08) |
| CFR12RRA473-- | .047 | 50 | F, G, J, K | CFR16RRA104-- | .10 | 50 | F, G, J, K | .095(2.41) | .245(6.22) | .295(7.49) | .200(5.08) |
| CFR12RRA563-- | .056 | 50 | F, G, J, K | CFR16RRA124-- | .12 | 50 | F, G, J, K | .095(2.41) | .295(7.49) | .295(7.49) | .200(5.08) |
| CFR12RRG223-- | .022 | 75 | F, G, J, K | CFR16RRA154-- | .15 | 50 | F, G, J, K | .095(2.41) | .295(7.49) | .295(7.49) | .200(5.08) |
| CFR12RRG273-- | .027 | 75 | F, G, J, K | CFR16RRA184-- | .18 | 50 | F, G, J, K | .095(2.41) | .295(7.49) | .295(7.49) | .200(5.08) |
| CFR12RRB103-- | .010 | 100 | F, G, J, K | CFR16RRA224-- | .22 | 50 | F, G, J, K | .095(2.41) | .395(10.03) | .395(10.03) | .300(7.62) |
| CFR12RRB123-- | .012 | 100 | F, G, J, K | CFR16RRA274-- | .27 | 50 | F, G, J, K | .095(2.41) | .395(10.03) | .395(10.03) | .300(7.62) |
| CFR12RRB153-- | .015 | 100 | F, G, J, K | CFR16RRA334-- | .33 | 50 | F, G, J, K | .095(2.41) | .395(10.03) | .395(10.03) | .300(7.62) |
| CFR12RRB183-- | .018 | 100 | F, G, J, K | CFR16RRA394-- | .39 | 50 | F, G, J, K | .095(2.41) | .395(10.03) | .395(10.03) | .300(7.62) |
| CFR12RRH682-- | .0068 | 150 | F, G, J, K | CFR16RRA474-- | .47 | 50 | F, G, J, K | .145(3.68) | .395(10.03) | .395(10.03) | .300(7.62) |
| CFR12RRH822-- | .0082 | 150 | F, G, J, K | CFR16RRA564-- | .56 | 50 | F, G, J, K | .145(3.68) | .395(10.03) | .395(10.03) | .300(7.62) |
| CFR12RRC392-- | .0039 | 200 | F, G, J, K | CFR16RRA684-- | .68 | 50 | F, G, J, K | .195(4.95) | .395(10.03) | .395(10.03) | .300(7.62) |
| CFR12RRC472-- | .0047 | 200 | F, G, J, K | CFR16RRA824-- | .82 | 50 | F, G, J, K | .195(4.95) | .395(10.03) | .395(10.03) | .300(7.62) |
| CFR12RRC562-- | .0056 | 200 | F, G, J, K | CFR16RRA105-- | 1.00 | 50 | F, G, J, K | .195(4.95) | .395(10.03) | .395(10.03) | .300(7.62) |
| CFR12RRK102-- | .0010 | 250 | F, G, J, K | | | | | | | | |
| CFR12RRK122-- | .0012 | 250 | F, G, J, K | | | | | | | | |
| CFR12RRK152-- | .0015 | 250 | F, G, J, K | | | | | | | | |
| CFR12RRK182-- | .0018 | 250 | F, G, J, K | | | | | | | | |
| CFR12RRK222-- | .0022 | 250 | F, G, J, K | | | | | | | | |
| CFR12RRK272-- | .0027 | 250 | F, G, J, K | | | | | | | | |
| CFR12RRK332-- | .0033 | 250 | F, G, J, K | | | | | | | | |

1/ APPLICATION NOTE. Care must be taken during wave soldering to assure that the CFR12 type capacitors are not damaged due to overheating. During the wave soldering operation, the CFR12 type capacitor shall not be exposed to conditions in excess of the resistance to solder heat requirements of MIL-C-55514.

2/ The complete type designation will include additional capacitance tolerance and failure rate level (M, P, R, or S).

Ⓓ TABLE II. Substitutability data.

| From | | To | From | | To |
|---|---|---|---|---|---|
| MIL-C-55514/8(ER) and MIL-C-55514/8A | MIL-C-55514/8B and MIL-C-55514/8C | MIL-C-55514/8D | MIL-C-55514/8(ER) and MIL-C-55514/8A | MIL-C-55514/8B and MIL-C-55514/8C | MIL-C-55514/8D |
| Type designation 1/ | Type designation 1/ | Type designation 1/ | Type designation 1/ | Type designation 1/ | Type designation 1/ |
| CFR12RRJ102-- | CFR12RRK102-- | CFR12RRK102-- | CFR12RRG102-- | CFR12RRK102-- | CFR12RRK102-- |
| CFR12RRJ122-- | CFR12RRK122-- | CFR12RRK122-- | CFR12RRG122-- | CFR12RRK122-- | CFR12RRK122-- |
| CFR12RRJ182-- | CFR12RRK182-- | CFR12RRK182-- | CFR12RRG182-- | CFR12RRK182-- | CFR12RRK182-- |
| CFR12RRJ222-- | CFR12RRK222-- | CFR12RRK222-- | CFR12RRG222-- | CFR12RRK222-- | CFR12RRK222-- |
| CFR12RRJ272-- | CFR12RRK272-- | CFR12RRK272-- | CFR12RRG272-- | CFR12RRK272-- | CFR12RRK272-- |
| CFR12RRJ332-- | CFR12RRK332-- | CFR12RRK332-- | CFR12RRG332-- | CFR12RRK332-- | CFR12RRK332-- |
| CFR12RRJ392-- | CFR12RRC392-- | CFR12RRC392-- | CFR12RRG392-- | CFR12RRC392-- | CFR12RRC392-- |
| CFR12RRJ472-- | CFR12RRC472-- | CFR12RRC473-- | CFR12RRG472-- | CFR12RRC472-- | CFR12RRC472-- |
| CFR12RRJ562-- | CFR12RRC562-- | CFR12RRC562-- | CFR12RRG562-- | CFR12RRC562-- | CFR12RRC562-- |
| CFR12RRJ682-- | CFR12RRH682-- | CFR12RRH682-- | CFR12RRG682-- | CFR12RRH682-- | CFR12RRH682-- |
| CFR12RRJ822-- | CFR12RRH822-- | CFR12RRH822-- | CFR12RRG822-- | CFR12RRH822-- | CFR12RRH822-- |
| CFR12RRJ103-- | CFR12RRB103-- | CFR12RRB103-- | CFR12RRG103-- | CFR12RRB103-- | CFR12RRB103-- |
| CFR12RRJ123-- | CFR12RRB123-- | CFR12RRB123-- | CFR12RRG123-- | CFR12RRB123-- | CFR12RRB123-- |
| CFR12RRJ153-- | CFR12RRB153-- | CFR12RRB153-- | CFR12RRG153-- | CFR12RRB153-- | CFR12RRB153-- |
| CFR12RRJ183-- | CFR12RRB183-- | CFR12RRB183-- | CFR12RRG183-- | CFR12RRB183-- | CFR12RRB183-- |
| CFR12RRJ223-- | CFR12RRG223-- | CFR12RRG223-- | CFR12RRB102-- | CFR12RRK102-- | CFR12RRK102-- |
| CFR12RRJ273-- | CFR12RRB273-- | CFR12RRB273-- | CFR12RRB122-- | CFR12RRK122-- | CFR12RRK122-- |
| CFR12RRJ333-- | CFR12RRA333-- | CFR12RRA333-- | CFR12RRB182-- | CFR12RRK182-- | CFR12RRK182-- |
| CFR12RRJ393-- | CFR12RRA393-- | CFR12RRA393-- | CFR12RRB222-- | CFR12RRK222-- | CFR12RRK222-- |
| CFR12RRJ473-- | CFR12RRA473-- | CFR12RRA473-- | CFR12RRB272-- | CFR12RRK272-- | CFR12RRK272-- |
| CFR12RRJ563-- | CFR12RRA563-- | CFR12RRA563-- | CFR12RRB332-- | CFR12RRK332-- | CFR12RRK332-- |
| CFR12RRJ683-- | CFR12RRA683-- | CFR16RRA683-- | CFR12RRB392-- | CFR12RRC392-- | CFR12RRC392-- |
| CFR12RRJ823-- | CFR12RRA823-- | CFR16RRA823-- | CFR12RRB472-- | CFR12RRC472-- | CFR12RRC472-- |
| CFR12RRJ104-- | CFR16RRA104-- | CFR16RRA104-- | CFR12RRB562-- | CFR12RRC562-- | CFR12RRC562-- |
| CFR12RRA102-- | CFR12RRK102-- | CFR12RRK102-- | CFR12RRB682-- | CFR12RRH682-- | CFR12RRH682-- |
| CFR12RRA122-- | CFR12RRK122-- | CFR12RRK122-- | CFR12RRB822-- | CFR12RRH822-- | CFR12RRH822-- |
| CFR12RRA182-- | CFR12RRK182-- | CFR12RRK182-- | CFR12RRH102-- | CFR12RRK102-- | CFR12RRK102-- |
| CFR12RRA222-- | CFR12RRK222-- | CFR12RRK222-- | CFR12RRH122-- | CFR12RRK122-- | CFR12RRK122-- |
| CFR12RRA272-- | CFR12RRK272-- | CFR12RRK272-- | CFR12RRH182-- | CFR12RRK182-- | CFR12RRK182-- |
| CFR12RRA332-- | CFR12RRK332-- | CFR12RRK332-- | CFR12RRH222-- | CFR12RRK222-- | CFR12RRK222-- |
| CFR12RRA392-- | CFR12RRC392-- | CFR12RRC392-- | CFR12RRH272-- | CFR12RRK272-- | CFR12RRK272-- |
| CFR12RRA472-- | CFR12RRC472-- | CFR12RRC472-- | CFR12RRH332-- | CFR12RRK332-- | CFR12RRK332-- |
| CFR12RRA562-- | CFR12RRC562-- | CFR12RRC562-- | CFR12RRH392-- | CFR12RRC392-- | CFR12RRC392-- |
| CFR12RRA682-- | CFR12RRH682-- | CFR12RRH682-- | CFR12RRH472-- | CFR12RRC472-- | CFR12RRC472-- |
| CFR12RRA822-- | CFR12RRH822-- | CFR12RRH822-- | CFR12RRH562-- | CFR12RRC562-- | CFR12RRC562-- |
| CFR12RRA103-- | CFR12RRB103-- | CFR12RRB103-- | CFR12RRC102-- | CFR12RRK102-- | CFR12RRK102-- |
| CFR12RRA123-- | CFR12RRB123-- | CFR12RRB123-- | CFR12RRC122-- | CFR12RRK122-- | CFR12RRK122-- |
| CFR12RRA153-- | CFR12RRB153-- | CFR12RRB153-- | CFR12RRC182-- | CFR12RRK182-- | CFR12RRK182-- |
| CFR12RRA183-- | CFR12RRB183-- | CFR12RRB183-- | CFR12RRC222-- | CFR12RRK222-- | CFR12RRK222-- |
| CFR12RRA223-- | CFR12RRG223-- | CFR12RRG223-- | CFR12RRC272-- | CFR12RRK272-- | CFR12RRK272-- |
| CFR12RRA273-- | CFR12RRG273-- | CFR12RRG273-- | CFR12RRC332-- | CFR12RRK332-- | CFR12RRK332-- |

1/ The complete type designation will include additional capacitance tolerance and failure rate level (M, P, R, or S).



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